

PARAS SHARMA

Ph.D. Student

School of Computing and Information,
Learning Research & Development Center,
University of Pittsburgh

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EDUCATION

Ph.D. in Computer Science

University of Pittsburgh
Advisor: Dr. Erin Walker

Aug. 2022 – Present

GPA:3.92/4

Master of Science in Computer Science

University of Southern California

Aug. 2019 – May 2021

GPA:3.86/4

Bachelor of Technology (with Hons.) in Computer Science & Engineering

Indian Institute of Technology (ISM), Dhanbad

Jul. 2013 – May 2017

GPA:9.55/10

RESEARCH INTERESTS

Educational Technology; Multimodal Learning Analytics; Dialogue Systems; Collaborative Learning;
Human-Computer Interaction; Large Language Models; Learner Modeling and Adaptation

RESEARCH EXPERIENCE

Graduate Student Researcher, Learning Research & Development Center

University of Pittsburgh

Aug. 2022 – Present

Pittsburgh, PA

- Developing HMMs to understand knowledge construction processes in human-human-robot learning interactions.
- Using LLMs to build theory-informed and responsive dialogue systems for learner reflections.
- Understanding lexical alignment trajectories in teachable human-human-robot interactions
- Implementing structure and surface gestures in a teachable robotic agent to study the impact of gestures in math learning.
- Working on building a Simulated Student System to emulate learner interactions in open-ended learning environments.
- Used HMMs and Sequence Mapping to model learner activity data and extract SRL and cognitive states that learners transition through in open-design settings.
- Designed a multimodal sensing system to detect learners' sensors, programming, and dialogue activities while building a robot.

Research Intern, Institute for Creative Technologies

University of Southern California

March 2021 – Aug. 2021

Los Angeles, CA

- Developed and tested different question-answering classification techniques on the Covid-Answers dataset for building a virtual mentor.
- Used Transformers to model the similarity between input questions and answers for the mentor.

Graduate Research Assistant, Social Media Analytics Lab

Keck School of Medicine, USC

Feb. 2021 – May 2021

Los Angeles, CA

- Developed framework to fetch and analyze video advertisement data of tobacco products on YouTube and TikTok.
- Created and analyzed various metrics for the impact of tobacco product advertisements by various companies using collected video data.

Graduate Research Assistant, Viterbi iPodia

University of Southern California

Feb. 2020 – Dec. 2020

Los Angeles, CA

- Developed diverse grouping algorithm to increase cultural and intellectual diversity among cohorts.
- Designed and implemented the backend of the framework, returning groups of students.
- Collected video and transcript data for student interactions grouped using the above algorithm.

TEACHING EXPERIENCE

Teaching Fellow, School of Computing & Information

University of Pittsburgh

Aug. 2024 – Dec. 2024

Pittsburgh, PA

- Worked as the instructor for CS0011 Introduction to Computing for Scientists.
- Developed and implemented lesson plans, projects, and exams.

INDUSTRY EXPERIENCE

Software Development Engineer, EC2

Amazon Web Services

June. 2021 – July 2022

Seattle, WA

- Developed an SDK to ease the onboarding of various workloads in Launch Wizard Service.
- Integrated Launch Wizard with Alpine service to better monitor the impact of service and API failures on customers.
- Improved tagging of Launch Wizard deployed workloads to enhance the revenue generation calculations.

Software Development Engineer Intern, EC2

Amazon Web Services

May 2020 – Aug. 2020

Seattle, WA

- Designed, developed, and deployed a dedicated host tenancy feature for SQL HA in the AWS Launch Wizard Service.
- Enabled Host Resource Group Integration for the AWS Launch Wizard Service.

Senior Software Engineer, Intelligence & IoT Division

Samsung R&D Institute

June 2017 – July 2019

Bangalore, India

- Implemented reinforcement learning in a personalized Social Robot using user-interaction feedback.
- Created the Context Engine Module and reinforcement learning module of Social Robot.
- Developed and deployed Device Identity and Locksmith Microservice on AWS EC2 with data stored in DynamoDB enabling IoT devices' access to the IoT cloud utilized by partners Vodafone and Sercomm.
- Designed and implemented Android as a Thing SDK to easily onboard any Android device onto the IoT cloud and expose its functionalities as controllable OCF resources.

Software Engineer Intern, Intelligence & IoT Division

Samsung R&D Institute

May 2016 – July 2016

Bangalore, India

- Developed anomaly Sensing and recommendation Engine for Smart IoT using IoTivity.
- Built and trained a neural network model for anomaly detection using the Samsung IoT device data.

PUBLICATIONS

- [1] **Paras Sharma**, YuePing Sha, Janet Shufor Bih Epse Fofang, Brayden Yan, Jess A. Turner, Nicole Balay, Hubert O. Asare, Angela E.B. Stewart, and Erin Walker. 2026. Hybrid LLM-Embedded Dialogue Agents for Learner Reflection: Designing Responsive and Theory-Driven Interactions. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI'26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3772318.3791582>
- [2] **Sharma, P.** et al. (2025). Beyond Static Measures: Temporal Analysis of Lexical Alignment in Human-Human Learning With a Teachable Robot. In: Cristea, A.I., Walker, E., Lu, Y., Santos, O.C., Isotani, S. (eds) *Artificial Intelligence in Education. AIED 2025. Lecture Notes in Computer Science()*, vol 15880. Springer, Cham. https://doi.org/10.1007/978-3-031-98459-4_20
- [3] Asano, Y., Litman, D., **Sharma, P.**, Fritsch, D., King-Shepard, Q., Nokes-Malach, T., . . . Walker, E. (2025). Multi-party Lexical Alignment in Collaborative Learning with a Teachable Robot. In A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds), *Artificial Intelligence in Education* (pp. 123–131). Cham: Springer Nature Switzerland.
- [4] Stewart, A. E. B., Solyst, J., Bao, X., **Sharma, P.**, Buddemeyer, A., Nkrumah, T., . . . Walker, E. (2025). Who's Got the Power? Data Feminism as a Lens for Designing AIED Engagement Systems. In A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds), *Artificial Intelligence in Education* (pp. 233–247). Cham: Springer Nature Switzerland.
- [5] Yuya Asano, Sabit Hassan, **Paras Sharma**, Anthony B. Sicilia, Katherine Atwell, Diane Litman, and Malihe Alikhani. 2025. Contextual ASR Error Handling with LLMs Augmentation for Goal-Oriented Conversational AI. In *Proceedings of the 31st International Conference on Computational Linguistics: Industry Track*, pages 374–386, Abu Dhabi, UAE. Association for Computational Linguistics. <https://aclanthology.org/2025.coling-industry.32/>
- [6] **Sharma, P.**, Bella, V., E. B. Stewart, A., Walker, E. (2024). Multimodal Sensing of Goals and Activities During Interactions with a Co-created Robot. In: Ferreira Mello, R., Rummel, N., Jivet, I., Pishtari, G., Ruipérez Valiente, J.A. (eds) *Technology Enhanced Learning for Inclusive and Equitable Quality Education. EC-TEL 2024. Lecture Notes in Computer Science*, vol 15160. Springer, Cham. https://doi.org/10.1007/978-3-031-72312-4_22
- [7] **Paras Sharma**, Angela E.B. Stewart, Qichang Li, Krit Ravichander, & Erin Walker. (2024). Building Learner Activity Models From Log Data Using Sequence Mapping and Hidden Markov Models. *Proceedings of the 17th International Conference on Educational Data Mining*, 584–593. <https://doi.org/10.5281/zenodo.12729890>
- [8] **Paras Sharma**, & Qichang Li. (2024). Designing Simulated Students to Emulate Learner Activity Data in an Open-Ended Learning Environment. *Proceedings of the 17th International Conference on Educational Data Mining*, 986–989. <https://doi.org/10.5281/zenodo.12730023>
- [9] Anthony Sicilia, Yuya Asano, Katherine Atwell, Qi Cheng, Dipunj Gupta, Sabit Hassan, Mert Inan, Jennifer Nwogu, **Paras Sharma**, Malihe Alikhani. “ISABEL: An Inclusive and Collaborative Task-Oriented Dialogue System”. In *Alexa Prize TaskBot Challenge 2 Proceedings*, 2023. <https://www.amazon.science/alex-prize/proceedings/isabel-an-inclusive-and-collaborative-task-oriented-dialogue-system>

HONORS AND AWARDS

3rd Position, Alexa Prize TaskBot Challenge 2. Amazon. \$50,000. 2023

Orrin E. and Margaret M. Taulbee Graduate Award for Excellence in Computer Science. Department of Computer Science, School of Computing and Information, University of Pittsburgh. 2025.

Pre-Doctoral Fellowship. School of Computing and Information, University of Pittsburgh. 2022-2023.

Viterbi Graduate Student Scholarship. Viterbi School of Engineering, University of Southern California. \$7,500. 2019.

Director's Award for Academic Excellence. Indian Institute of Technology (Indian School of Mines), Dhanbad. 2015.

Travel Scholarship. 17th Educational Data Mining Conference (EDM2024), Atlanta, Georgia. \$2600. 2024

Travel Scholarship. 25th International Conference on Artificial Intelligence in Education (AIED 2024), Recife, Brazil. \$1200. 2024

Travel Scholarship. 26th International Conference on Artificial Intelligence in Education (AIED 2025), Palermo, Italy. \$1350. 2025

MENTORSHIP EXPERIENCE

Daniel Fritsch. Sep 2024 - Present, BS Computer Science, University of Pittsburgh

Project: Determining Lexical Alignment Trajectories and Knowledge Construction Processes in a teachable human-human-robot setting.

Michael Puthumana. Sep 2024 - Present, BS Computer Science, University of Pittsburgh

Project: Hardware modifications to support external speaker and microphone sensors in the Social programmable robotic kits.

Krit Ravichander. Jan 2024 - Aug 2025, BS Computer Science and Psychology, University of Pittsburgh

Project: Building structure and surface gestures for the NAO robot.

Brayden Yan. Jan 2025 - Aug 2025, BS Computer Science, University of Pittsburgh

Project: Updating dialogue systems with LLMs to promote reflections in interactions between learners and their robot designs.

Travis Mindel. Jan 2025 - Apr 2025, BS Information Science and Psychology, University of Pittsburgh

Project: Using Media Pipe to build automated gesture replication in a NAO robot.

Jude Gilligan. Jan 2025 - Apr 2025, BS Computer Science, University of Pittsburgh

Project: Integrating real-time Lexical Alignment into the teachable robot interface.

Siyena Gezahegn. Sep 2024 - Apr 2025, BS Computer Science, University of Pittsburgh

Project: Hardware modifications to support external speaker and microphone sensors in the Social programmable robotic kits.

Zeana El Hajomar. Sep 2024 - Jan 2025, BS Computer Science, University of Pittsburgh

Project: Interface and UX Improvements for Teachable Robot.

Audrey Flood. May 2024 - July 2024, FACET Lab Summer Intern, University of Pittsburgh

Project: UI/UX design for the frontend interface of the social Programmable Robots project.

Qichang Li. Jan 2024 - Apr 2024, MS Computer Science, University of Pittsburgh.

Project: Development of a Dialogue Agent by Distilling Knowledge from a Large Language Model for Robot Programming Assistance.

Raquel Buege. Jan 2024 - Apr 2024, BS Computer Science, University of Pittsburgh

Project: Frontend implementation of the social programmable robots interface.

Veronica Bella. Jan 2023 - May 2024, BS Electrical Engineering, University of Pittsburgh

Project: Custom hardware extensions to the hummingbird robotics kit to sense immediate addition/removal of hardware components.

UNIVERSITY SERVICE

LRDC Executive Committee. University of Pittsburgh. Sep 2024 - Present

Computer Science Graduate Student Organization (President). University of Pittsburgh. Sep 2024 - Present

VOLUNTEER SERVICE

Student Volunteer. 26th International Conference on Artificial Intelligence in Education (AIED 2025)

Student Volunteer. 25th International Conference on Artificial Intelligence in Education (AIED 2024)

TECHNICAL SKILLS

Research: Prototyping, Quantitative and Qualitative methods, Statistical Testing, Experimental Design, user modeling, Data Mining

Languages: Python, Java, C/C++

Libraries: PyTorch, TensorFlow, NumPy, pandas, Matplotlib, NLTK

Frameworks/Tools: Frontend and Backend Development, Git, Android Development, AWS EC2, Elastic Search, IBM Watson, AWS S3